FCK-U SYSTEM

Frame Cab Kinetic Utilization

Automaker Technical Brief — Open Source Prior Art Disclosure

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Open Source Technical Brief	[Your Name]	March 3, 2026	Public Domain — No Patents

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Executive Summary

The FCK-U System is a comprehensive vehicle safety and comfort architecture addressing three unresolved problems in current passenger vehicle design:

Problem	Current State	FCK-U Solution
Vibration transmission to occupants	Seat-level attenuation only; cabin rigidly coupled to frame	Tri-stage cooperative adaptive suspension with CG-pivoted floating cabin
Crash energy transmission to occupants	Crumple zone only; cabin decelerates at frame rate	Omnidirectional pyrotechnic cabin displacement away from impact vector
EV traction battery fire risk	Battery structurally integrated into occupant floor	Physical cabin-battery separation at impact; fire-barrier skid plate

The system is architected specifically for compatibility with steer-by-wire and drive-by-wire platforms (Cybertruck, Rivian R1T, and similar), as these architectures eliminate the mechanical constraints that would otherwise prevent cabin displacement.

Module 1: Tri-Stage Cooperative Adaptive Suspension

1.1 Architecture

Three independently controllable vibration isolation stages share a real-time CAN bus. A Conflict Resolver algorithm running at ≥1 kHz enforces alternating stage dominance to prevent inter-stage resonance amplification — the primary failure mode of stacked passive isolation systems.

Stage	Component	Frequency Domain	Energy Function
1 — E-CESA	ECU-controlled electromagnetic dampers, frame-mounted	0.5–5 Hz (macro: potholes, body roll, dive)	Primary regen loop → main traction battery (est. 100–500W/corner)
2 — Yoke Struts	Dual-angled linear generators, V-config at CG pivot	15–150 Hz (micro: road texture, gravel, drivetrain NVH)	Secondary regen loop → isolated Cabin Survival Battery (12V/48V LFP)
3 — Active Pivot	Gimbaled bearing, variable resistance all axes	All frequencies — rotational/torsional	No energy function; conflict resolution and crash-mode release

1.2 Center-of-Gravity Swivel Pivot

The pivot must track the dynamic CG of the loaded cabin. Static CG:

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z_cg = (m_structure·z_structure + m_occupants·z_occupants + m_battery·z_battery) / m_total
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A motorized vertical adjustment mechanism with ±80mm travel compensates for occupant load variation (0–5 passengers). Lateral CG shift is handled by differential compliance bias in the Yoke struts without physical pivot relocation.

1.3 Predictive / Self-Learning Control

- GPS-tagged road surface frequency profiles stored per route; suspension pre-conditioned 200–500ms before known events
- Differential WSS data used to predict curve entry; predictive cabin lean applied to cancel centrifugal force on occupants
- Occupant load model refined over time via accumulated CG measurement data

1.4 Comparison to Existing Systems

System	Vibration Isolation Point	Cabin Decoupled?	Predictive?	Energy Regen?
Passive coil/spring	Wheel-to-frame	No	No	No
Semi-active (MagneRide)	Wheel-to-frame	No	Limited	No
Active (Bose Suspension)	Wheel-to-frame	No	Yes	No
Audi eROT	Wheel-to-frame	No	No	Yes (primary)
FCK-U (this disclosure)	Frame-to-cabin	Yes	Yes	Yes (dual-loop)

Module 2: Active Kinetic Displacement — Omnidirectional Breakaway

2.1 System Overview

Replaces the conventional rigid cabin-to-frame mounting with four independently addressable pyrotechnic mounts at the corners of the cabin-skid plate assembly. Upon confirmed unavoidable impact, the Cabin ECU selects a firing sequence based on the impact azimuth vector derived from forward sensor data.

2.2 Force Requirements

For a cabin mass m ≈ 600 kg and a representative crash deceleration of 25g (245 m/s²):

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F_required = m × a = 600 kg × 245 m/s² = 147,000 N (33,060 lbf)
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This force profile is consistent with automotive-grade pyrotechnic linear actuators. Electric actuators are excluded due to insufficient force density and response time. Pyrotechnic technology is mature and certified in existing automotive safety applications (seatbelt pretensioners, hood actuators, battery disconnects).

2.3 Firing Sequence Logic

Impact Azimuth	Mount Firing Sequence	Resulting Cabin Vector	Primary Benefit
0° (direct frontal)	All 4, simultaneous	Rearward	Maximum crumple zone extension
1–89° (left offset)	Left mounts T+0, right T+10ms	Rearward-right	Angular escape from impact corner
271–359° (right offset)	Right mounts T+0, left T+10ms	Rearward-left	Angular escape from impact corner
180° (rear impact)	All 4, simultaneous	Forward	Battery separation priority
Rollover (>45° pitch/roll)	All 4, simultaneous	Ground plane drop	Skid plate ground contact

2.4 Compatibility Note

> *This system requires steer-by-wire and brake-by-wire architectures. Vehicles with mechanical steering columns or conventional hydraulic brake boosters physically cannot implement cabin displacement. This is why the FCK-U architecture is specifically targeted at next-generation EV platforms where these mechanical constraints are already absent.*

Module 3: Composite Fire-Barrier Skid Plate

3.1 Material Stack

Layer	Material	Thickness	Primary Function
Outer (ground-facing)	Ceramic-coated carbon fiber composite	≥8mm	Fire barrier (rated ≥1100°C continuous), sliding surface
Core	Closed-cell polyisocyanurate foam	≥40mm	Structural stiffness, thermal insulation, positive buoyancy
Inner (cabin-facing)	3mm aluminum honeycomb panel	3mm	Impact energy distribution, secondary structure

3.2 Structural Role Post-Separation

In current EV platforms, the traction battery pack provides structural contribution to the floor system. Upon cabin separation, the skid plate must assume this structural role independently. The composite laminate specification is designed to provide equivalent torsional and bending stiffness to a conventional vehicle floor pan.

3.3 Multi-Function Summary

- Structural floor replacement post-separation
- Thermal barrier (ceramic outer layer prevents thermal runaway propagation to occupant compartment)
- Sliding surface (low-friction ceramic enables predictable deceleration on asphalt, gravel, wet surfaces)
- Emergency flotation (sealed foam core provides positive buoyancy for full loaded cabin assembly)

Module 4: Cabin-Battery Physical Separation Interface

4.1 Interface Architecture

A defined Separation Interface Plane (SIP) replaces the conventional integrated battery-floor structure. All services crossing the SIP are equipped with crash-triggered disconnects:

Service	Disconnect Type	Fire Order	Post-Separation State
High-voltage traction circuit	Pyrotechnic contactor (existing Tesla tech)	First (T+0ms)	HV bus de-energized before any movement
Thermal management coolant loops	Spring-loaded auto-seal valve	Second (T+5ms)	Both sides sealed, no coolant spray
Low-voltage / data bus (CAN, Ethernet)	Blind-mate quick-disconnect	Third (T+10ms)	60GHz wireless assumes control authority

Service	Disconnect Type	Fire Order	Post-Separation State
Mechanical cabin mounts	Pyrotechnic shear bolts	Last (T+20ms)	Cabin free to displace on skid plate

> *The sequencing of HV disconnect before mechanical separation is critical. Arc flash at the moment of mechanical disconnect is the primary fire ignition risk at the SIP. This risk is eliminated by ensuring HV isolation precedes movement by a minimum of 15ms.*

4.2 Post-Separation Battery Management

The traction battery remains with the vehicle frame after cabin separation. The frame-resident Main ECU continues monitoring and management:

- Immediate: Contactor confirmed open, HV bus confirmed de-energized
- T+100ms: Cooling pump at maximum flow rate for thermal management
- T+500ms: Frame-side emergency beacon transmitting GPS, crash severity, battery thermal state
- Continuous: Thermal runaway detection active; if threshold exceeded, secondary suppression system activates

Module 5: Federated Dual-ECU Architecture

5.1 System States

ECU	Normal Operation	Pre-Impact (T-500ms)	Post-Separation
Main ECU (frame-resident)	Full authority — motors, battery, steering, braking	Initiates stiffening protocol; primes pyro circuits	Battery management, thermal monitoring, frame beacon
Cabin ECU (cabin-resident)	Warm standby — mirrors steering/brake state; charges survival battery	Assumes authority over cabin systems; arms airbags	Full authority — steer-by-wire, brake-by-wire, doors, windows, cabin beacon

5.2 Cabin ECU Power Budget

Load	Peak Draw	Continuous Draw	Duration Required
Steer-by-wire actuators	200W	80W	Until stop (~60s)
Brake-by-wire	150W	40W	Until stop

Load	Peak Draw	Continuous Draw	Duration Required
Emergency lighting	30W	30W	30 min
60GHz wireless	15W	15W	Until stop + 5 min
GPS / SOS beacon	10W	5W	2 hours minimum
Airbag standby	5W	5W	30 min
Total	410W peak	175W continuous	—

A 2 kWh LFP cell (approx. 10kg, motorcycle-battery form factor) provides full post-separation functionality with substantial margin. LFP chemistry selected for superior thermal stability and reduced thermal runaway risk versus NMC/NCA.

Module 6: W-FCK-U Wireless Control Redundancy

6.1 Latency Analysis

Required: End-to-end steer/brake command latency <50ms (human perception threshold at any road speed).

Layer	Technology	Latency	Notes
RF propagation	60GHz (IEEE 802.11ad/WiGig)	~1ms	Sub-meter cabin-to-frame distance, line-of-sight
Protocol overhead	WiGig MAC/PHY	5–8ms	Deterministic scheduling mode required
Actuator response	Electric rack / brake actuator	15–20ms	Platform-dependent
Total	—	21–29ms	Compliant with <50ms requirement

6.2 Antenna Diversity Requirement

60GHz is susceptible to metallic occlusion and water absorption. Minimum antenna array:

- Cabin side: 3 nodes — front-center, rear-left, rear-right
- Frame side: 3 nodes — front-left, front-right, rear-center
- Diversity switching: Automatic, <1ms switchover, ensures ≥1 LOS pair at any cabin displacement angle up to 45°

Module 7: Implementation Pathway for EV Platforms

7.1 Required Platform Prerequisites

Prerequisite	Status on Cybertruck	Status on Rivian R1T	Notes
Steer-by-wire	Yes (confirmed)	No (conventional EPS)	Required — cannot implement without
Brake-by-wire	Yes (confirmed)	Partial	Required for post-separation braking
Forward LIDAR/RADAR	Yes (Vision + radar)	Yes	Required for pre-impact trigger
Pyro battery disconnect	Yes (standard Tesla)	Yes	Existing tech, requires extension
OTA software updates	Yes	Yes	Required for ECU handoff logic

7.2 Novel Engineering Requirements

- Structurally independent cabin floor (replace battery-structural contribution with skid plate composite)
- Defined Separation Interface Plane with sequenced multi-service disconnects
- CG-tracking pivot assembly with ±80mm vertical adjustment
- Cabin-resident LFP survival battery with isolated charging loop
- 60GHz antenna array on both cabin and frame sides of SIP

Closing Statement

> *All concepts in this document are released to the public domain. No patent application has been filed or will be filed. This publication constitutes prior art. Any manufacturer may implement these systems immediately, at no cost, with no licensing requirement.*

The originator's sole intent is the acceleration of these safety technologies into production vehicles. Questions and engineering collaboration are welcomed.

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